

Spoorthy Reddy Merugu

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EDUCATION

Bachelors in Electronics and Communication Engineering

CMR College of Engineering & Technology

Aug 2017 – May 2021
Hyderabad, India

- **Courses:** SQL, JAVA Programming, Data Structures And Algorithm, Control Systems, VLSI Design, Embedded Systems, Real Time OS, Digital Image Processing.
- **CGPA:** 9.42/10.00

Masters in Computer Science

University of Alabama at Birmingham

Jan 2023 – present
Birmingham, United States

- **Courses:** Deep Learning, Database Systems, Cloud Computing, Adv Web Application Dev, Probability & Statistics in CS, Structured query language.
- **CGPA:** 3.80/4.00

PROFESSIONAL EXPERIENCE

FactSet Research Systems INC.

SOFTWARE ENGINEER

Aug 2021 – Dec 2022
Hyderabad, India

- Developed comprehensive testing strategies and conducted rigorous testing processes to ensure robustness and reliability of software solutions; integrated automated testing tools which reduced manual testing efforts by 50%.
- Processed and delivered real-time financial news stories from various vendors in the form of JSON, XML, and so on to various clients.
- Implemented a Python service to lessen reliance on a Machine Learning service, improving tagging performance from 60% to 100%.
- Engineered innovative updates for subsystems of end-user applications software across various platforms, resulting in a 40% decrease in system downtime and a 25% improvement in user satisfaction metrics.
- Assisted in preparing monthly and quarterly credit reports for internal and external stakeholders, including Regulatory Reporting Schedules.
- Conducted second-level review and validation of key reports prepared by team members, ensuring accuracy and compliance.
- Collaborated with business partners and technology teams to identify risks and opportunities in the portfolio, addressing ad hoc reporting requests.
- Participated in working groups for bank-wide data enhancement projects, contributing to improved data quality and reporting efficiency.
- Supported the creation of presentations, training materials, and reporting tools in collaboration with managers, utilizing SharePoint and databases.
- Provided advice and guidance to junior analysts, fostering a collaborative and supportive team environment.
- Technologies used included Python, C++, MySQL, Git, Perforce, Shell Scripting, UNIX, and Unit Test, Statistical Analysis Systems (SAS), Structured Query Language (SQL), Power Business Intelligence (Power BI).

PROJECTS

Enhancing Regulatory Reporting with Data Visualization

- Revolutionized monthly and quarterly regulatory submissions by spearheading a project focused on optimizing reporting processes through advanced data visualization techniques.
- Gathered diverse data from internal databases, spreadsheets, and external sources, resulting in a 30% reduction in system latency.
- Leveraged SAS, SQL, and Python to cleanse, transform, and centralize data, ensuring 99.9% accuracy and integrity. Utilized Tableau, Power BI, and Python libraries to craft interactive visualizations, offering insights into regulatory metrics for compliance.
- Engineered automated scripts to detect and rectify discrepancies in data, contributing to a 25% increase in data quality. Contributed ideas for bank-wide data enhancements in working groups, evaluating reporting processes and visualizations for optimization and refinement.
- **Technologies:** SAS, SQL, Python for data integration and transformation, Tableau, Power BI, and Python libraries (Matplotlib, Seaborn) for visualization, and Microsoft Office suite for communication and documentation.
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Online Ticket Booking System

Web Interface

May 2023 – Jul 2023

- Provided a user-friendly interface, seamless booking process by improving efficiency, and enhancing customer satisfaction.
- Engineered a dynamic partnership with developers, UI/UX designers, and database administrators to fuse various elements of the ticket booking system, leading to a 30% reduction in system latency and enhanced user experience.
- Collaborated on the refinement of user authentication using JSON Web Tokens (JWT) to ensure secure user interactions and data protection.
- **Front-end:** HTML, CSS, JavaScript, React.js
- **Back-end:** Node.js, Express.js, Django, Flask
- **Database:** MongoDB, SQL Server.

Cloud-Based File Sharing Tool

May 2023 – Jul 2023

- Developed user-friendly web interface for secure file uploads and sharing, utilizing AWS EC2 for hosting and ensuring smooth interactions.
- Crafted robust file upload capabilities leveraging the AWS S3 SDK, with DynamoDB managing file metadata and user interactions.
- Automated email notifications via AWS Lambda and SES for improved engagement.
- Ensured seamless user experience through testing, including reliable email notifications and precise database operations.
- **Key Technologies:** Amazon EC2, Amazon S3, AWS Lambda, Amazon SES, Amazon DynamoDB, HTML, CSS, JavaScript

Face Recognition System Based on Raspberry Pi Using Python

Jan 2021

- Developed an image capture system that stores video images as two-dimensional arrays in the database; integrated facial recognition to verify the identity of individuals, reducing unauthorized access incidents by 40%
- **Technologies:** RASPBERRY-PI and PYTHON

CERTIFICATES

Big Data and Data Analytics — By DELLEMC CERTIFICATION • **Introduction to Internet of Things And Embedded Systems** — By Coursera • **Microsoft Power BI Data Analyst** — Used Power BI for data connections, prepared Excel data, created reports and dashboards, completed a capstone project, and prepared for the Microsoft PL-300 exam, gaining skills in Data Analysis, Excel, SQL, Power BI, and Power Query. • **Coursera Data Analysis and Visualization with Power BI** — I have learned to add and design visualizations for accessible reports and dashboards, and perform data analysis. Skills gained: Data Analysis, Data Visualization, Power BI, Report Design, and Dashboard Design. • **C and DS workshop** — By CMRCET Coding club. • **Python Course Certificate** — By NPTEL and Coursera.

SKILLS

Languages

PYTHON, JAVA, Azure, Angular, HTML/CSS, SQL, JavaScript

Database

SQL Server, Oracle, PostgreSQL.

Tools

Power BI, Tableau, Probability and Statistics, Data Analytics, Pandas, Scikitlearn, NumPy, Matplotlib, Git, Perforce, UNIX, AWS, EC2, Kubernetes, Elastic search, EBS.